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Iberian Triassic paleomagnetism revisited: Intraplate block rotations versus polar wandering

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Abstract. A paleomagnetic study of over 220 samples, obtained from 21 sites from Triassic redbeds in the Algarve (S Portugal) and La Meseta (SE Spain), yields two paleomagnetic poles at lat = 62°N,lon = 202°E, A95 = 9 and lat=54°N, lon=178°E, A95=7 respectively. The Algarve paleopole reflects a rotation associated with the Alpine evolution of the Iberian Atlantic margin. A comparison of the La Meseta pole with other data indicates a progressive clockwise rotation of the Eurasian Plate from the Late Paleozoic to Middle Triassic, followed by rapid drift until the Upper Triassic. This late Triassic drift event may be related to the opening and spreading of the Tethys towards the west.

Introduction

The precise reconstruction of the latest Paleozoic and early Mesozoic Pangea is still being debated. Four different proposals have been forwarded (Van der Voo et al., 1976; Morel and Irving, 1981; Smith et al., 1980; Hallam, 1983), some of them implying large megashear between Laurasia and Gondwana. Central to solving this debate is the reconstruction of the pre-fragmentation paleogeography of the Pangea supercontinent. Paleomagnetism provides an independent source of data addressed to this problem. Geological evidence usually comes from the borders of the cratons, but paleomagnetic investigation tends to focus on the stable areas. Mesozoic rocks of the Iberian Peninsula have been studied paleomagnetically for a long time. This interest arises from a) the peculiar position of Iberia as a plate between the Atlantic and the Mediterranean and between two megaplates (Eurasia and Africa) to the north and south respectively; b) the rotation that described Iberia's motion during the opening of the Gulf of Biscay associated with the NE motion of the African Plate. All these factors instigated numerous paleomagnetists to focus on the Triassic remanence directions of Iberia and compare them to the expected or predicted directions for Eurasia and Gondwana (see Van der Voo, 1993 for a recent review). Thus, once Clegg et al. (1957) claimed evidence for a counterclockwise rotation of the Iberian Peninsula on the basis of paleomagnetic results from Triassic rocks from Iberia and Eurasia, numerous studies were addressed to the establishment of reliable Mesozoic paleopoles to document for such crustal shifting (Van der Lingen, 1960; Schwarz, 1963; Van der Voo, 1967, 1969; Van Dongen, 1967; Van der Voo and Zijderveld, 1971; Stauffer and Tarling, 1971; Van der Voo and Boessenkool, 1973; Vandenberg, 1980; Schott and Peres, 1987; Parés et al., 1988; Galdeano et al., 1989).

The Triassic is selected above other Mesozoic periods due to 1) the generalized paleomagnetic stability of almost all Triassic rocks: nearly 70 % of the materials composing the Triassic period are suitable redbeds -continental siliciclastic rocks; 2) since the earliest studies it was recognized that the opening of the Gulf of Biscay, and thus the rotation of Iberia, took place after Triassic times.

The Triassic segment of the apparent polar wander path (APWP) is still controversial: nearly all of the Triassic poles come from the Pyrenean belt, and the remaining poles are obtained from allochthonous terraines, although Schott (1985) showed that these poles fit reasonably with the kinematic evolution models for Iberia. On the other hand, only one Triassic locality that is stable and undeformed has been paleomagnetically studied (Van der Voo, 1967). The resulting paleopole was not considered in the APWP elaboration by Schott (1985), who argued that the sampling was restricted to one small area and that magnetic cleaning was incomplete. We believe that two factors - a) intraplate block rotations and b) continental drifting-, can account for the observed differences in the Triassic directions of the Iberian Plate. In this paper we report on new paleomagnetic data from the Iberian Triassic which address directly the above problem.

Geological setting and sampling

In Western and Central Europe the Triassic was a period of lithosphere extension that resulted in the formation of a complex network of grabens and troughs (Vegas and Banda, 1982, Ziegler, 1982). The infilling deposits are related to changes in sea-level, rates of subsidence and/or rates of terrigenous input (Busson, 1982; Ziegler, 1982). A cyclicity in the Triassic sedimentation is expressed through an alternation of dominantly carbonate sequences and siliciclastic and/or evaporite sequences. The Triassic sediments constitute the beginning of the Alpine cycle sedimentation almost all over the Iberian Peninsula. The vertical succession is composed of a red siliciclastic lower layer, a carbonatic intermediate layer and an evaporitic upper layer. This succession can be considered as "Germanic" in character. In detail, however, there are some complexities, concerning lithofacies as well as biofacies.

Our sampling was designed to complement that of Van der Voo (1967): two sampling areas were selected (Algarve and SE Iberian Massif), the locations of which are shown in Fig. 1. The sampling areas were chosen so as to represent the southeast and southwest extremes of the Iberian Massif, and in both cases the Triassic deposits overlap the Paleozoic basement that constitutes the Iberian stable ("cratonic type") area. Nine sites were sampled in the Algarve region (ALG), along the southern coast of Portugal, where the Triassic red sandstones dip gently to moderately southward and unconformably overlie the folded Carboniferous rocks. Similarly, along the SE border of Iberian Massif or the so called Meseta region (MET), Triassic flat lying red beds rest on folded Ordovician metapelites. Twelve sites were sampled in this region.

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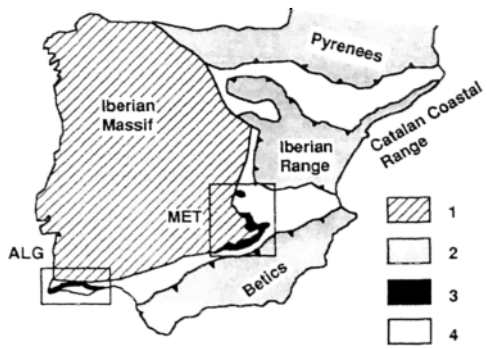


Figure 1. General geologic map of the Iberian Peninsula depicting the main structural units: 1- Paleozoic Iberian Massif; 2- Tertiary Apine thrust-and-fold belts; 3- Triassic outcrops of the Iberian Massif domain; 4- Main Cenozoic basins.

The selection of the sampling area is based on the premise that the Triassic deposits remained rigidly attached to the Iberian Massif during the Alpine deformation as evidenced by field observations. A precise chronology cannot be completely established due to the scarcity of data. Nevertheless, Palain (1968), on the basis of palinoflora data, reports an Upper Triassic age for the Algarve red beds. Sopeña et al. (1983) argue for an Upper Triassic age for the SE Iberian Massif deposits. Thus at present, without a detailed Triassic chronology being possible, the fossiliferous findings reveal that collectivly, the sampled sediments from the Iberian Peninsula span the Upper Triassic period.

Paleomagnetic results and discussion

Paleomagnetic analysis has been conducted on over 220 samples using a double shielded GM400 CCl 3-axes cryogenic magnetometer at the Institute of Earth Sciences in Barcelona. This magnetometer has a noise level of $5x10^{-0.6}$ A/m, well below the intensity values of the studied samples. Thermal demagnetization performed with a TSD-1 oven (Schonstedt), with at least 10 steps from room temperature to the limit of reproducible results, was used throughout (Fig. 2). Eventually, eight successive alternating field steps were applied before thermal treatment. Characteristic remanent magnetization (ChRM) directions were calculated by principal component analysis (Kirschvink, 1980). To obtain a site mean direction, Fisher's statistics were used to average ChRM directions (Fig. 3). In most cases there is a good within-site grouping of the data (mean α_{95}=16, Table I). Two sites with extremely large cones of confidence (greater than 25) were not used in the computation of the mean direction.

Although no fold test was available, the following observations provide confidence that the calculated ChRM directions are accurate and primary in age: (1) the presence of geomagnetic reversals, (2) the mean directions that differ significantly from the present geomagnetic field before bedding correction and (3) the high magnetic stability of the analyzed redbeds (Fig. 2). Our new results support those obtained by Van der Voo (1967), reaffirming the northward-directed magnetization of La Meseta Triassic rocks. The statistical parameters for each of the mean directions -ALG and MET- allow us to estimate a significant angular difference of 15 degrees between both Triassic domains (Fig. 3). Consequently the ALG and MET paleopoles are either not representative of the same Triassic paleofield or indicate intraplate block rotations or a combination of both.

The critical assessment of the available geological and paleomagnetic data by Van der Voo (1993) provides some clues on the angular difference observed between the Algarve and the SE Iberian Massif Triassic red beds: Van der Voo noticed that the paleomagnetic poles for Europe and North America during the Triassic period are not fixed, but there is a change in both paleolongitude and paleolatitude of the poles between the Upper Permian and Early Jurassic, describing a cusp at about 200 Ma, which is well defined in the North American APWP (Gordon et

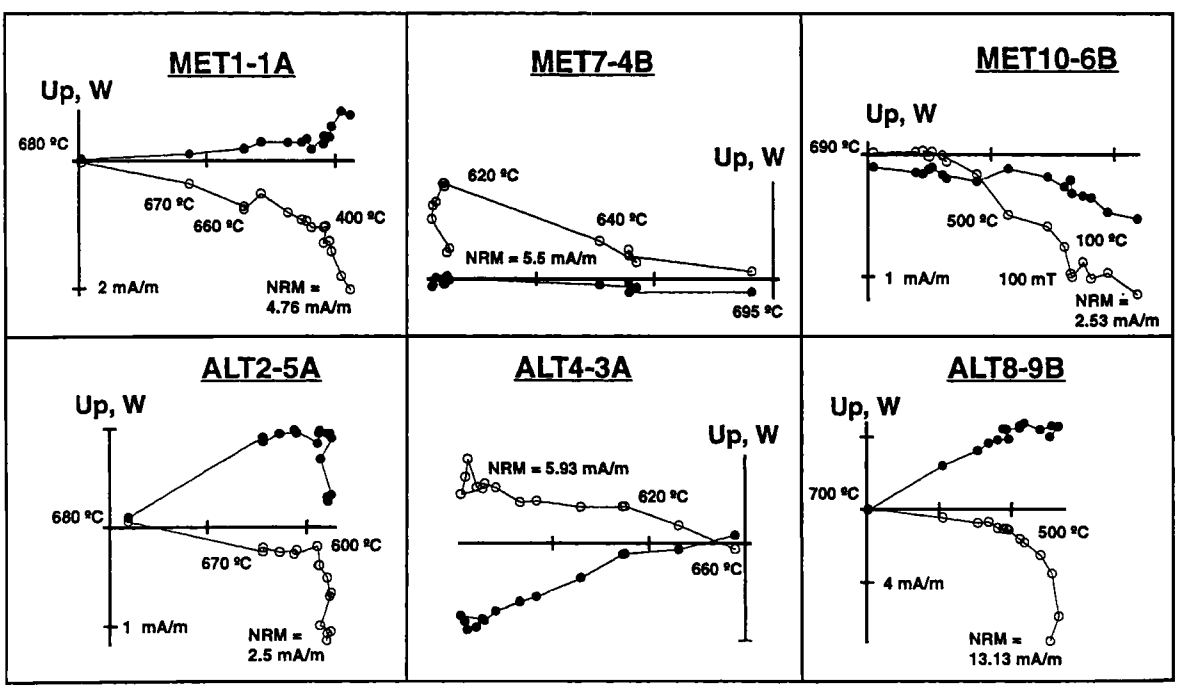


Figure 2. Typical thermal demagnetization diagrams of Triassic red sandstones (full circles- horizontal component; open circles- vertical component). Alternating fields up to 100 mT were applied to sample MET10-6B before thermal treatment, as shown in the plot. All projections are tectonic corrected.

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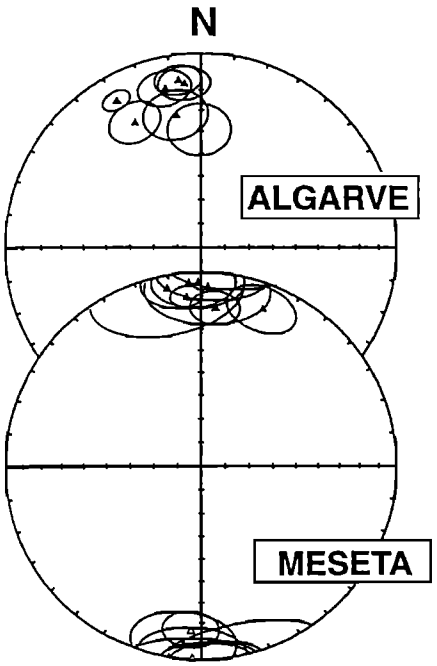


Figure 3. Equal-area projection of the mean directions of magnetization obtained from each site, with 95% confidence circles. Full symbols: downward magnetization directions. Open symbols: upward magnetization directions. Site mean directions are listed on Table 1.

al., 1984; Ekstrand and Butler, 1989). The data illustrating this feature is shown in Fig. 4. The dark shaded curve shows the declinations and paleolatitudes for Iberia, as calculated from the European reference poles, whereas the light shaded curve shows the observed declination and paleolatitude for Iberian Triassic localities (Van der Voo, 1993). The shifting between both curves -extrapolated and observed- in Fig. 4 is due to the Cretaceous rotation of Iberia relative to Eurasia as discussed above. Thus both curves would nearly overlap by rotating the one on the left by 35 degrees clockwise, i.e. closing the Bay of Biscay. As they stand on Fig. 4, it can be seen that declinations and inclinations of both extrapolated and observed curves are in very good agreement: The path swept out by the directions follows a regular clockwise progression, from west to east, and a sharp increase in paleolatitude at Middle-Upper Triassic times.

The paleomagnetic directions found in the Algarve and the SE Iberian Massif are plotted on Fig. 4 as circle and square respectively. The position of the Triassic pole from MET seems to fit reasonably on the curve of the observed data in Iberian Peninsula. Thus these results agree with the kinematic evolution of the European plate. However, the pole position from the Algarve Triassic sites (ALG) lies westwards of the calculated curve. Moreover, the inclination of this pole implies a higher latitude than that of the MET pole, possibly reflecting a younger age of the Algarve Triassic deposits. Thus, we conclude that Alpine intraplate deformation has produced the observed discrepancies between the ALG and MET poles. Assuming that the MET reflects the apparent polar wander path according to Fig. 4, the Algarve region would then have rotated counterclockwise to produce the observed declination. The higher paleolatitude value of the ALG pole and in general of the Middle-Upper Triassic poles can be explained as a consequence of the northward drift of Tethys during the early Mesozoic.

Concluding remarks

The Triassic deposits from the SE Iberian Massif are undeformed and can be regarded as a Mesozoic tegument over the Paleozoic crystalline massif. Although in the Algarve the Triassic deposits unconformably overlie the Paleozoic basement, some deformation is evidenced by moderate tilting of bedding. Taken at face value, an eastward displacement of the poles as well as a sharp increase in paleolatitude would appear to be a reasonable

Table. 1. Paleomagnetic data from Algarve and La Meseta

Site	Site lat	Site long	Db	Ib	Da	Ia	C/S	N+R	k	α_{95}
ALT1	37° 12'	352° 30'	357	-1	359	39	11/11	10	16.8	12.1
ALT2	37° 10'	352° 18'	333	-22	332	28	16/19	14+2	15.7	9.6
ALT4	37° 10'	351° 30'	347	-6	347	17	14/19	11+4	18.5	9.1
ALT5	37° 15'	351° 45'	352	-7	352	14	15/13	10	39.6	7.8
ALT7	37° 05'	351° 10'	354	21	349	31	13/15	11	15.7	11.9
ALT8	37° 04'	351° 05'	331	0	330	14	15/19	17	59.0	4.7
ALT9	36° 59'	351° 04'	355	5	354	16	13/16	15	19.8	8.8
Algarve (ALT) paleomagnetic pole: lat = 62°N, lon = 202°E, A95 = 9										
MET1	38° 12'	356° 30'	355	13	355	13	10/9	9	90.7	5.4
MET2	38° 15'	356° 50'	349	7	349	7	19/16	6+1	7.1	24.4
MET3	38° 18'	356° 55'	186	-8	186	-8	8/15	7	13.8	16.9
MET4*	38° 51'	357° 20'	163	14	163	14	9/13	3	16.8	31.0
MET5	38° 40'	357° 25'	2	8	2	8	10/11	9	7.7	19.9
MET6*	38° 40'	357° 30'	162	22	162	22	11/10	10	4.2	27.0
MET7*	38° 36'	357° 11'	184	-16	184	-16	8/7	4	95.0	9.5
MET8	38° 42'	357° 04'	193	-1	193	-1	9/8	8	12.2	16.5
MET9	38° 42'	357° 01'	5	19	5	19	9/8	6	62.2	8.6
MET10	39° 24'	356° 45'	359	5	359	5	13/8	6+1	18.6	14.4
MET11	39° 24'	356° 45'	21	14	21	14	9/8	5	42.9	11.8
MET12	39° 24'	356° 45'	356	5	356	5	7/5	5	28.1	14.7
La Meseta (MET) paleomagnetic pole: lat = 54°N, lon = 178°E, A95 = 7										

Site: Site number; Site lat: Site latitude; Site long: Site longitude; Db: Declination before bedding correction; Ib, Inclination before bedding correction; Da: Declination after bedding correction; Ia, Inclination after bedding correction; C/S: Number of cores obtained in the field and number of specimens used in statistics respectively; N+R: Number of normal and reversed directions, k. Fisher's precision parameter; α_{95}: half angle of the 95% confifence cone.

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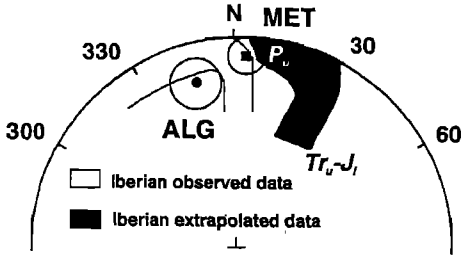


Figure 4, Equal-angle projection of observed (light shaded curve) and extrapolated (dark shaded curve) declinations and paleolatitudes for Iberian data. The extrapolated/predicted values are derived from the stable European paleopoles (Van der Voo, 1993). Algarve (ALG) and SE Iberian Massif (MET) mean paleopoles are plotted as a circle and square with their respective confidence cone. The curves span the time from upper Permian (P_u) to lower Jurassic (J_l).

explanation for the Triassic segment of the Iberian APWP. Alpine deformation related to the evolution of the Atlantic margin is a plausible mechanism to interpret the pole observed for the Algarve region.

Thus, the Triassic paleomagnetism of Iberia revisited confirms the paleopole computed by Van der Voo (1969), and collectively the data emphasize the progressive, moderate clockwise rotation of the Eurasian plate during the latest Paleozoic and early Mesozoic. A sharp increase in latitude during the Triassic period is also evidenced from the Iberian data, revealing a rapid northward motion of the Tethys realm. This major drift event is contemporaneous with the first Mesozoic rift stage described for the Iberian domain by Sopeña et al. (1988) and evaluated by Salas and Casas (1993) and thereby may have been related to the opening and spreading of the Tethys towards the west.

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